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(54) Title: FORMULATION OF A TURMERIC EXTRACT COMPRISING CURCUMINOIDS, METHOD OF PRODUCTION THEREOF, THE USE OF THE FORMULATION AND PRODUCTS COMPRISING THEREOF

(57) Abstract: The invention provides a formulation of a turmeric extract comprising curcuminoids, dispersible in water, and the method of production of the turmeric extract formulation dispersible in water, the use thereof in food products, feed products, medicinal products, beverage and beverage concentrate, dietary supplement, pharmaceutical composition, preferably activating the metabolism of acetaldehyde, acting as anti-inflammatory agent as well as relates to products comprising formulation, beverages and their concentrates, food products, feed products, dietary supplements, medicinal products, pharmaceutical composition and uses thereof.

Formulation of a turmeric extract comprising curcuminoids, method of production thereof, the use of the formulation and products comprising thereof

DESCRIPTION

TECHNICAL FIELD

The invention provides a formulation of a turmeric extract comprising curcuminoids, dispersible in water, and the method of production of the turmeric extract formulation dispersible in water, the use thereof in food products, feed products, dietary supplements, medicinal products, beverage and beverage concentrate, dietary supplement, pharmaceutical composition, preferably activating the metabolism of acetaldehyde and products comprising them, beverages and their concentrates, food products, feed products, dietary supplements, medicinal products, pharmaceutical composition and uses thereof.

STATE OF THE ART

Curcuma longa, commonly called turmeric, is a tropical plant of the *Zingiberaceae* family originating in southern Asia. Turmeric rhizomes have traditionally been used as a coloring ingredient, but also as a spice. Turmeric is quite commonly used as a spice, but also as an additive to food or beverages (Govindarajan, 1980; Salzer et al., 1975). Dried turmeric rhizomes comprise 3-5% of essential oils and 0.02-6.0% of curcuminoids, such as curcumin, demethoxycurcumin and bisdemethoxycurcumin. These curcuminoids may occur in various amounts and proportions relative to each other, what depends on the biological and environmental factors associated with the cultivation of the rhizome itself, and in the case of oleoresin or extracts, on the technology of obtaining them. Apart from the coloring and spice properties, turmeric, and especially the curcuminoids comprised in it, exhibit various healing properties confirmed by pharmacological studies conducted in various animals. Turmeric extracts show anti-inflammatory, anti-fungal and

anticancer effects (Amalraj et al., 2017; Yamamoto et al., 1997; Apisariyakul et al., 1995; Ruby et al., 1995). Curcuminoids, in order to affect living organisms (humans and animals) must first be delivered to the body, and then be absorbed from the digestive tract into the bloodstream (Amalraj et al., 2017; Jäger et al., 2014).

Curcuminoids are compounds that do not dissolve in water and, moreover, their solubility in vegetable and animal fats is very limited and does not exceed 1 mg ml^{-1}, what significantly hinders the bioavailability of curcuminoids from the gastrointestinal tract. Curcuminoids best dissolve in organic solvents such as acetone, followed by ethyl acetate, methanol and ethanol (in the case of ethanol, to increase the solubility of curcuminoids it is required to heat the solution to the boiling temperature, i.e. about 78 °C, which causes thermal decomposition of curcuminoids) (Amalraj et al., 2017; Anderson et al., 2000). Due to the toxicity of the solvents, solutions obtained in this form cannot be administered to living organisms. Methods of dissolving or dispersing curcuminoids in water are sought, so that they can be introduced into the body of humans or animals in this form. Studies have shown that the physical or chemical form of curcuminoids plays a key role in their bioavailability. It was shown that dispersion of curcuminoids in an aqueous environment significantly improves their bioavailability (Jäger et al., 2014). In addition, the purpose of developing a water-soluble or dispersing curcuminoids formulation is to facilitate their combination with an appropriate flavor composition, so that they can be consumed by humans and animals much more readily. Curcuminoids in concentrated form, like the turmeric powder itself, have a very intense, characteristic taste and are bitter, which practically disqualifies the possibility of consuming them directly. In this form, they can be either encapsulated or used as a food additive. Development of curcuminoids formulation enabling their dispersion in the aqueous environment is a big technological challenge, also due to the low solubility of curcuminoids in fats (fat solubility <1 mg g^{-1}, whereas in water curcuminoids are completely insoluble). In the case of products intended for food, the formation of a dispersion system must be carried out with organic solvents approved for use in food production, which must then be removed from the final product. Moreover, their content in the final product must meet the legal requirements for the approximation of the laws of the Member

States on the extraction solvents used in the production of foodstuffs and food ingredients (Directive of the European Parliament and Council 2009/32/WE of April 23rd, 2009).

In the world literature, many methods for the preparation of curcuminoids formulations dispersible in water are described. A very common way to obtain curcuminoids dispersible in water is to dissolve them in 70-80% ethanol together with β-cyclodextrin, followed by long heating and mixing for at least 4 h at the temperature of 70 °C (Mangolim et al., 2014; Marcolino et al., 2011). Solvents (ethanol and water) are usually removed by spray drying, but also as a result of vacuum evaporation or freeze drying. This method of obtaining curcuminoids dispersible in water has several disadvantages. The first, basic, are the unfavorable chemical transformations of curcuminoids in the water-ethanol environment associated with their thermal decomposition. Research by Wang et al. (1999) showed that curcuminoids in an aqueous environment under the influence of elevated temperature, even at the most favorable pH of the environment (curcuminoids show the highest stability at a pH between 3 and 6) undergo decomposition. Wang et al., (1999) showed that heating curcumin at 37 °C for 2 hours causes as much as 10% loss of this compound. The formulation of curcuminoids dispersible in water should be carried out at the lowest possible temperatures, especially in the process of joining them with the aqueous phase. An additional disadvantage of the formulation using β-cyclodextrin is its limited use in food production. β-cyclodextrin (E 459) can be used for the production of, e.g., beverages, but its content cannot be greater than 1 g kg^{-1} (Commission Regulation (EU) no. 1129/2011). This limits the possibility of introducing any amount of curcuminoids into the final product.

Other formulations of curcuminoids dispersible in water are these compounds adsorbed on silicates, e.g., silicon dioxide (Martins et al., 2013). The use of curcuminoids adsorbed on silicon dioxide (E 551) in the production of, e.g., beverages is quite limited, because such a formulation is not thermodynamically stable and such particles quickly undergo sedimentation. This is due to the high density of silicon dioxide, as well as the curcuminoids bound to it themselves.

There are many different curcuminoid formulations available on the world market, but not all of them are suitable for the production of beverages, that is their dispersion in water. Meriva® produced by Indena SpA (Italy) consists of curcuminoids in the amount of 18-20%

adsorbed on microcrystalline cellulose with the addition of soy lecithin (Cuomo et al., 2011). This product is more suitable for the preparation of solid preparations or as an additive to foods, but other than beverages due to the lack of solubility of microcrystalline cellulose, which falls out of the solution and undergoes sedimentation. LongVida® produced by Verdure Sciences (USA) is a preparation that comprises approx. 20% of curcuminoids suspended in fat particles, which are solid at room temperature, with the addition of phosphatidylcholine (Gota et al., 2010). The preparation is generally used for the production of dietary supplements in the form of tablets or capsules, it can also be added to various foods, however, it is not suitable for permanent dispersion of curcuminoids in water. CurQfen® produced by Akay Flavors & Aromatics Pvt. Ltd., MicroActive Curcumin produced by BioActives LLC (USA) comprises approx. 25% of curcuminoids suspended in short-chain triacylglycerols with the addition of free fatty acids and polyglycerol esters. The formulation is stabilized with hydroxypropyl methylcellulose, sodium alginate and microcrystalline cellulose (Madhavi et al., 2014). This formulation was mainly developed for the production of tablets/capsules comprising in its composition a highly bioavailable curcumin. The Micronized Curcumin preparation, produced by Raps GmbH & Co., KG (Germany), comprises in its composition approx. 25% of curcumin powder dispersed in triacetin (glycerol triacetate) and a mixture of lipids obtained from the seeds of soybean, oil palm and rapeseed (Schiborr et al. 2014). The preparation is insoluble in water and is not suitable for obtaining stable curcuminoids dispersible in water. NovaSOL® produced by FRUTAROM from Israel is a micellar curcumin in a concentration of approx. 7% in Tween-80 (Schiborr et al., 2014). The preparation cannot be used in food, because Tween-80 is not approved for use in food in many countries, including the EU. CurcuWin® produced by OmniActive Health Technologies (India) comprises in its composition 20-28% of turmeric extract (a mixture of three curcuminoids), which is suspended in a mixture with the composition 63-75% of polyvinylpyrrolidone (E 1201), 10-40% cellulose derivatives with the addition of 1-3% of natural antioxidants (Jager et al., 2014). The preparation enables the production of tablets/capsules, but is not suitable for the production of beverages. CURCUGREEN® (BCM-95®) produced by Arjuna Natural Extracts Ltd. (India) is an essential oil consisting of 45% ar-turmerone and curcuminoids (Antony et al., 2008). The CURCUGREEN® preparation is insoluble in water and is not suitable for the production of beverages. Curcumin

C3 Complex® + Bioperine® produced by Sabinsa USA is a mixture of piperine and curcuminoids without a special formulation (Shoba et al., 1998). The preparation is not suitable for dispersion in water. CAVACURMIN® produced by Wacker Chemie AG (Germany) comprises approx. 15% curcuminoids in combination with γ-cyclodextrin (Puroura et al., 1998). This product disperses in water, however, at high temperatures curcuminoids form conjugates with cyclodextrins (this has been described earlier), and this promotes the thermal breakdown of curcuminoids. In addition, γ-cyclodextrin is not allowed for use in food in the EU and the preparation cannot be used for the production of foods including beverages. Theracurmin™ produced by Theravalues Corp. (Japan) is a nano-colloidal preparation comprising in its composition up to 30% of curcuminoids. The preparation additionally comprises glycerin and Ghatti gum (Sasaki et al., 2011). The product disperses well in water, it is suitable for the production of beverages, however, at the present moment it cannot be used for the production of food, because Ghatti gum is not approved for food production, as an additive in the EU. In addition, due to the presence of maltose and maltodextrin, the Theracurmin™ preparation is characterized by a sweet taste, which limits its use in food products. The content of maltose and maltodextrin in the Theracurmin™ preparation, due to their high glycemic index and high calorific value prevents its use in the production of low-calorie or special purpose beverages, e.g., for diabetics or overweight people.

According to the World Health Organization, alcohol (ethanol) is at the top of the list of risk factors threatening the health of the population. However, due to the widespread consent to its consumption, the scale of medical problems, caused mainly by the toxic effects of ethanol metabolites, is constantly growing. One of the best described substances, in terms of toxicology, generated during alcohol metabolism is acetaldehyde. Acetaldehyde is the first product of ethanol oxidation. It is a compound 30 times more toxic, mutagenic and carcinogenic than ethanol, with documented multi-organ activity, especially within the liver and lungs. The conversion reaction of ethanol to acetaldehyde is catalyzed by alcohol dehydrogenase (ADH), while the main enzyme catalyzing the oxidation of acetaldehyde to low-toxic acetic acid is aldehyde dehydrogenase (ALDH). To date, 19 ALDH isoenzymes encoded by different gene loci and showing a wide variety of substrate specificity have been identified. Despite the existence of many molecular forms of ALDH, only enzymes encoded by the ALDH1 and ALDH2 genes are

involved in the process of acetaldehyde degradation. Cytosolic ALDH1 is present in many tissues, including the brain, but the catalytic efficiency of this enzyme is low (Km approx. 50 µM). Much higher affinity for acetaldehyde is shown by mitochondrial ALDH2 (Km<5µM; pH 7,5), which occurs not only in hepatocytes, but also in other types of cells, e.g., gastric epithelium, glial and neurons. The loss of enzymatic activity of ALDH2 is the result of a one-point, nonsense mutation in the gene at position 1510, which results in the change of a base G to A in the exon 12. The activity of ALDH2 is closely related to racial affiliation: the mutation leading to the loss of enzymatic activity occurs mainly in Oriental populations (Japan, Taiwan, China, Korea, Hong Kong, Singapore, Vietnam, Malaysia etc. and it affects up to 50% of the population, while in the case of Caucasians, these values fluctuate in the range of 8-10%. The phenomenon of decreased ALDH2 activity also occurs in patients with liver cirrhosis, regardless of the etiology of the disease and the amount of alcohol consumed. The concentration of acetaldehyde in the blood, which is a derivative of the amount of alcohol consumed and the activity of alcohol dehydrogenase determining the rate of elimination of this compound, has a significant impact on the occurrence and course of many diseases of the liver, cardiovascular system, bronchial asthma, Alzheimer's disease and many cancers of the respiratory system and the gastrointestinal tract. Due to the very adverse biological effects of acetaldehyde, it is in the interest of public health to reduce its level in the human body as quickly and effectively as possible. On the dietary supplements and medical preparations market, anti-inflammatory preparations as well as preparations for supplementing the deficiency of vitamins, electrolytes and minerals with a symptomatic effect, alleviating the effects of alcohol and acetaldehyde poisoning, as well as few preventive preparations whose function is to support the work of the liver and accelerate the metabolism of alcohol and its metabolites are available, such as Morning Recovery (MoreLabs) or GETADAY (Drink6 Iberia S.L.U.) based on herbal extracts with the active substance DHM (dihydromyricetin) and herbs used in Ayurvedic medicine, i.a. ginseng and opuntia fig extract. The active substance in ReadyQ (Handok) is DHM and curcumin, and in the alcorythm® preparation (Nomi Biotech Corporation) – curcumin. Due to the low native bioavailability, despite the unequivocally positive effect of curcumin on the activation of acetaldehyde metabolism, the use of curcumin is limited, and in preparations containing curcumin, it is necessary to use substances that support

bioavailability, such as piperine, which significantly negatively affect the taste of the preparation and cause irritation of the mucosa in sensitive people. The lack of curcumin formulations dispersible in water, having increased bioavailability and high stability, based on ingredients approved for use in the food and pharmaceutical industries, characterized by a neutral taste and smell, is a serious technical limitation for the development of preventive preparations, including those activating the metabolism of acetaldehyde, based on the use of curcuminoids.

All the above described commercially available curcumin formulations were developed to maximize the bioavailability of curcuminoids, bypassing the technological aspects necessary to develop a formulation suitable for aqueous dispersion and, consequently, for the production of beverages.

DISCLOSURE OF THE INVENTION

TECHNOLOGICAL PROBLEM

A technological problem arising from the state of the art is the lack of available turmeric extract formulation in powder form dispersible in water, assuming that the turmeric extract comprises high concentrations of curcuminoids, preferably min. 90% w/w of curcuminoids, including curcumin, demethoxycurcumin and bisdemethoxycurcumin, the dispersion of which in water will be stable, without any observable, and noticeable for a long time, sedimentation or creaming process. After dissolution of the formulation in water, the system should be uniformly colored. Preferably, the turmeric extract content in the final formulation shall be at least 10%. All the substances used in the production of a formulation should be approved for use in foods, including, in the EU, the accordance with the current regulations in this area. The emulsification process should preferably take place at maximum the room temperature (20 °C ± 2 °C) to avoid thermal decomposition of curcuminoids. Higher temperature results in a decrease in viscosity, and thus an increase in the particle size distribution, which adversely affects the stability of the dispersion system.

AIM OF THE INVENTION

The aim of the present invention is to overcome the indicated inconveniences resulting from the state of the art. This aim was achieved by developing the formulation solution, as well as the method of production thereof and the product obtained by this method enabling the obtaining of a turmeric extract formulation dispersible in water with increased stability, increased homogeneity, wherein in the method of production of the curcumin formulation in the form of an emulsion solution and dry form, all substances used in the production process are approved for use in food production.

The inventors have unexpectedly found that the use of an unobvious system of excipients, preferably in the form of starch octenylsuccinate and cellulose derivatives as a carrier and the use of an organic solvent approved for use in food production, preferably in the form of ethyl acetate, methyl acetate, butyl acetate, ethanol, butanol, particularly preferably ethyl acetate, allows the formation of an emulsion with a narrow and low particle distribution in the homogenization process (the vast majority with a diameter of less than 1 µm, preferably in the range from 0.1 to 1 µm), what ensures the formation, in process of the spray drying, of a dispersible formulation comprising curcuminoids of turmeric extract with the desired physicochemical properties and increased bioavailability .

ESSENCE OF THE INVENTION

The innovative research carried out by the inventors showed that the use of a combination of starch octenylsuccinate and cellulose derivatives, in particular hydroxypropylcellulose, or hydroxypropyl methylcellulose as a carrier for the turmeric extract comprising curcuminoids, dissolved in an organic solvent approved for use in food production which is ethyl acetate, methyl acetate, butyl acetate, ethanol, butanol, particularly preferably ethyl acetate, allows the obtaining of an emulsion in the homogenization process. In addition, the inventors have found that the addition of a polysaccharide solution, preferably a high viscosity cellulose derivative, particularly preferably carboxymethyl cellulose (CMC), preferably the solvent is water, allows to stabilize the emulsion. The obtained emulsion is stable and has a narrow and low distribution of particles with a diameter less than 1 µm without visible sedimentation or creaming process, preferably for a period of not less than 12 months.

The combination of starch octenylsuccinate with modified cellulose as a carrier for turmeric extract dissolved in organic solvent, enables the production of turmeric extract formulation according to the invention in the form of an emulsion with a narrow and low particle distribution, even using only turbo-vortex homogenization. In order to obtain the turmeric extract formulation according to the invention in a dry form, most preferably powder, stabilization of the emulsion system during drying of the curcumin formulation in the form of an emulsion consisting in dilution of the produced emulsion in an aqueous solution of carboxymethyl cellulose (CMC) at the concentration of 0.001 to 1%, preferably from 0.01 to 0.5%, preferably 0.10 to 0.20%, preferably 0.1-1.4%, in a proportion of not less than 1:6, preferably in a proportion of about 1:8 and drying of the emulsion stabilized in this way to obtain a powder, e.g., by spray drying. The produced powder – curcumin formulation in powder form is still characterized by its narrow and low particle distribution, and after dissolution of the resulting powder, a stable dispersion system is obtained due to previous stabilization by carboxymethyl cellulose.

Surprisingly, it turned out that it is possible to obtain dispersed phase particles for a turmeric extract comprising curcuminoids with a dispersed phase particle size less of than 1 µm, what enables the obtaining of a system, stable from the point of view of thermodynamics of emulsion systems. The lower diameter range of the resulting particles does not affect the stability of the emulsion production, although obtaining most of the particles in a certain range, preferably in the particle diameter range of 0.1-1 µm, facilitates the industrial processing of such a uniform product at a later stage. With such a small particle size of the dispersed phase for the extract prepared in this way, the sedimentation speed calculated from the Stokes equation

$$V = [(2r^2 g(\rho_p - \rho_f)]/(9\eta)$$

will be low, which is exactly what the final curcumin formulation according to the invention provides.

More unexpectedly, it turned out that the curcumin formulation according to the invention shows higher bioavailability, i.e. it is better absorbed by the body than the known and available on the market preparations comprising curcumin, as well as it is characterized by a higher permeability through the intestinal cell layer. Unexpectedly, it turned out that the

curcumin formulation according to the invention provides improved properties in the regulation of liver function, it better activates the action of liver enzymes, and especially enzymes of the dehydrogenases group, including aldehyde dehydrogenase, it also exhibits greater anti-inflammatory, antitumor and antidepressant activity than the known and available curcumin preparations. The following explanations and general definitions refer to the descriptions of the preferred embodiments of the invention below.

The present invention, as illustrated below, may be suitably produced in the absence of any element or elements, restriction or restrictions, not specifically disclosed herein.

Whenever the term “comprising” or “including” is used in this description and claims, it does not exclude other elements. For the purposes of the present invention, the term “consisting of” is considered a preferred embodiment of the term “comprising”, “including”. If a group is defined below, which contains at least a number of embodiments, this should also be understood as disclosing a group, which preferably consists of only these embodiments.

The terms “about” or “approximately” in the context of the present invention stand for a range of accuracy that a person skilled in the art will understand to be sufficient to still provide the technical effect of the given feature. The term “typically” is to be understood as a deviation from the indicated numerical value of ± 10%, preferably ± 5%.

Technical terms are used according to their usual meanings. If a specific meaning is used, the meaning will be described in detail in relation to the context in which the term was used.

By stability is to be understood as the formation of a dispersed system with a dispersed phase particle size <1 μm. By the term “stability” is to be understood the formation of a dispersion system that is optically homogeneous, with no visible creaming or sedimentation process. The system is considered stable if during centrifuging in a centrifuge at 5000 rpm for 20 min and at the temperature of 25 °C, the amount of sediment at the bottom of the tube does not exceed 30% relative to the initial weight of the powder formulation dispersed in water.

Dispersion means a system composed of at least two immiscible phases, of which at least one is a finely divided system (dispersed phase), dispersed in the second phase, with continuous character (dispersive phase), called the dispersion medium.

Solubility means the ability of a solid substance to dissolve in the liquid phase and produce a dispersion system forming a heterogeneous mixture.

The invention relates to a curcumin formulation in a dry form, preferably powder, water-soluble, providing a dispersion system, in which the dispersed phase is characterized by the particle size of <1 μm. Additional features of the curcumin formulation are the following:

- The formulation is produced without the use of ethanol or other alcohols or ketones (e.g., acetone)
- The formulation is neutral in taste, that is: no sweet, sour, or bitter-burning taste, that is the characteristic turmeric flavor
- The formulation is produced at low temperatures, which favorably affects the chemical stability of curcuminoids.

The invention provides a curcumin formulation in a dry form comprising:

- starch octenylsuccinate in the amount of 0.1-45% w/w, more preferably 25-35% w/w, particularly preferably 28-32% w/w;
- a cellulose derivative in the amount of 0.01-20% w/w in the form of hydroxypropyl methylcellulose, hydroxypropylcellulose or a mixture thereof, preferably the cellulose derivative is in the amount of 5-15% w/w, particularly preferably 8-10% w/w;
- turmeric extract in the amount of 0.01-20% w/w, wherein the turmeric extract comprises at least 90% w/w of curcuminoids, including curcumin, demethoxycurcumin and bisdemethoxycurcumin, preferably the turmeric extract in the amount of 5-15% w/w, particularly preferably the turmeric extract in the amount of 8-10% w/w;
- β-glucan in the amount of 0.1-10% w/w, preferably 3-5% w/w, particularly preferably 3.8-4.2% w/w;
- a cellulose derivative with the viscosity of over 5600 mPas in the amount of 10-70% w/w, preferably in the amount of 30-60% w/w, particularly preferably in the amount of 48-52% w/w, wherein the cellulose derivative with the viscosity of over 5600 mPas is selected from carboxymethyl cellulose, methylcellulose, ethylcellulose, hydroxypropylcellulose, ethyl methyl cellulose, cross-linked cellulose gum, preferably, the cellulose derivative with the viscosity of over 5600 mPas is carboxymethyl cellulose;

wherein at least 90%, more preferably at least 95% of the particles, preferably at least 99% of the particles have a diameter of below 1 μm.

In the preferred curcumin formulation in a dry form at least 90%, more preferably at least 95% of the particles, preferably at least 99% of the particles have a diameter in the range from 0.1 to about 1 μm.

The invention also relates to the method of production of a curcumin formulation in a dry form, which comprises the following steps:

a) a turmeric extract with a minimum content of 90% of curcuminoids is dissolved in an organic solvent, approved for use in food production, preferably in ethyl acetate, methyl acetate, butyl acetate, ethanol, butanol, particularly preferably ethyl acetate;

b) the dissolved turmeric extract from step a) is added to a solution of a carrier, which is a mixture of polymers comprising starch octenylsuccinate and a cellulose derivative, preferably the cellulose derivative is hydroxypropylcellulose, hydroxypropyl methylcellulose, dissolved in a polar solvent, preferably in water, wherein preferably cooling is applied, wherein the temperature of the mixture of polymers during the addition of the dissolved turmeric extract falls within the temperature from about 5 °C to about 10 °C (the use of higher temperatures affects the viscosity of the continuous phase, what consequently contributes to obtaining an emulsion system with an unfavorable particle size distribution, i.e. with a diameter over 1 μm);

c) the mixture obtained in step b) is homogenized to obtain an emulsion, preferably at a temperature from about 5 °C to about 10 °C (the use of higher temperatures affects the viscosity of the continuous phase, what consequently contributes to obtaining an emulsion system with an unfavorable particle size distribution, i.e. with a diameter of over 1 μm);

d) the emulsion obtained in step c) is diluted with an aqueous solution of β-glucan (it is a polymer that positively affects the stability of the dispersion system), preferably, with the β-glucan concentration of 1-8%, preferably from oat or barley, with a low viscosity (preferably, with the viscosity in the range of 40 – 200 mPas) is further briefly homogenized;

e) the emulsion obtained in step d) is stabilized by combining it with a cellulose derivative with a high viscosity (> 5600 mPas), preferably a 0.1 to 1.4% aqueous solution of carboxymethyl cellulose, methylcellulose, ethylcellulose, hydroxypropylcellulose, ethyl methyl cellulose, cross-

linked cellulose gum, more preferably carboxymethyl cellulose with a high viscosity, by diluting, preferably, in a ratio of not less than 1:6, more preferably not less than 1:8, stirring it all mechanically;

f) the stabilized emulsion from step e) is subjected to drying to provide a curcumin formulation in a dry form, wherein over 90%, preferably over 95%, preferably over 99% of the obtained particles have a diameter size of below about 1 µm,

more preferably at least 90%, more preferably at least 95% of the particles, more preferably at least 99% of the particles have a diameter in the range from 0.1 to about 1 µm.

In the preferred method of production of a curcumin formulation, the organic solvent used to dissolve the turmeric extract in step a) is ethyl acetate, methyl acetate, butyl acetate, ethanol, butanol, more preferably ethyl acetate. In the preferred the method, the polar solvent used to dissolve the carrier in step b) is water.

In the preferred method of production of the curcumin formulation, the cellulose derivative used to produce the carrier is hydroxypropylcellulose, hydroxypropyl methylcellulose, or a mixture thereof,

wherein the preferred mixture of hydroxypropylcellulose to hydroxypropyl methylcellulose falls within the ratio of up to 10:1.

In the preferred method of production of the curcumin formulation, in step d), the emulsion is diluted with 1-8% β-glucan, preferably β-glucan from oat or barley, or a mixture thereof.

In the preferred method of production of the curcumin formulation, the cellulose derivative with a high viscosity is CMC, preferably in the concentration from 0.01 to 0.5%, more preferably in the concentration from 0.10 to 0.20%, more preferably in the concentration from 0.1 to 1.4% w/w.

In the preferred method of production of the curcumin formulation, the solution of the carrier is obtained by mixing polymers mechanically, wherein the mixing is preferably carried out for not less than 10 h, preferably about 12 h.

In the preferred method of production of the curcumin formulation, the solution of the carrier is dissolved in a polar solvent, in weight proportions from 1:5 to 1:25, preferably 1:10.

In the preferred method of production of the curcumin formulation, drying in step f) is carried out using spray drying with the temperature at the inlet of the dryer in the range of 110 – 160 °C, more preferably about 150 °C, and not more than 65 °C at the outlet, preferably not more than about 55 °C. It was unexpectedly found that the use of low temperatures in the range of 2-20 °C, preferably from about 5 °C to about 15 °C, more preferably from about 5°C to about 10 °C, more preferably about 7 °C or in a water bath, preferably an ice bath in stages b) and c), enables the production of an emulsion with the desired particle size range. Higher temperatures affect the viscosity of the continuous phase, what consequently contributes to obtaining an emulsion system with an unfavorable particle size distribution with a diameter over 1 µm.

In the preferred embodiments of the method of production of the curcumin formulation in a dry form intended to provide a stabilized emulsion characterized by a narrow and small distribution of particles in the diameter range from about 0.1 to about 1 µm, without a noticeable sedimentation or creaming process, a polymer carrier of starch octenylsuccinate and a cellulose derivative is used. The carrier is produced from a mixture of starch octenylsuccinate and a cellulose derivative, preferably with the viscosity in the range of 40 – 200 mPas, particularily preferably hydroxypropylcellulose, hydroxypropyl methylcellulose, preferably in the proportions of 0.1 to 30 g of starch octenylsuccinate (e.g., PURITY GUM 2000®) to 0.02 to 20 g of hydroxypropylcellulose or hydroxypropyl methylcellulose, or a mixture thereof, in the proportions from 0:0 to 10:1. After mechanical mixing of the polymers, they are dissolved in a polar solvent, which is particularly preferably water, in weight proportions from 1:5 to 1:25, preferably 1:10. The obtained mixture is mechanically mixed to dissolve and combine polymers, preferably for the proper dissolution and dissolve and combining of the polymers, for not less than 10 h, preferably about 12 h.

Separately, a solution from turmeric with a minimum content of 90% of curcuminoids is prepared in an organic solvent approved for use in food production, preferably ethyl acetate. The solution is obtained by dissolution of turmeric extract (min. 90% of curcuminoids) in the proportion of 0.01 to 3 g in an organic solvent approved for use in food production, preferably ethyl acetate in the amount from 0.19 to 60 ml. The dissolved curcuminoids extract is added to a mixture of polymers (proportion of the polymers 2:3 w/w) and homogenized using, e.g., a

mechanical homogenizer. Homogenization can be carried out in a variety of ways, continuously, for example by gradually increasing the rotation speed, but most preferably it is carried out in two steps, by first producing a pre-emulsion and then carrying out the proper homogenization. The two-step production of the emulsion is carried out, for example, at a rotational speed in the range of 5 000 rpm – 15 000 rpm, preferably at a rotational speed of the working element of 10,000 rpm until the pre-emulsion is formed. The entire mixture, from the moment the curcuminoids are introduced into the mixture of polymers, is cooled, for example in an ice water bath. The temperature of the polymer mixture at the moment of introduction of the extract dissolved in ethyl acetate, preferably should not exceed about 10 °C and preferably should not be lower than about 5 °C. After the production of the pre-emulsion, it is subjected to proper homogenization using a homogenizer, preferably a turbo-vortex homogenizer and process parameters in the range of 20 000-24 000 rpm, preferably 22 000 rpm for 3 minutes, after which time water or an aqueous solution of β-glucan is added, preferably a 1-8% aqueous solution of β-glucan, preferably from oat or barley with a low viscosity (about 40-200 mPas) in an amount from 0.5 to 210 ml (in the proportion of 1:0.25 w/w) and it is further homogenized, for example, for 1 min. The emulsion prepared in this way is immediately combined with a polysaccharide solution, preferably a cellulose derivative with a high viscosity, preferably a 0.1 – 1.4% aqueous solution of carboxymethyl cellulose (CMC) with a high viscosity. This technological treatment enables the production of a stable formulation, that is one with the particle size distribution of the dispersed phase <1 μm in diameter (for at least 90% of the particles, preferably 95%, most preferably for over 99% of the particles of the dispersed phase). The emulsion is diluted with water in a proportion of not less than 1:6, more preferably not less than 1:8, stirring it all mechanically using a stirrer, preferably a propeller stirrer, and the stabilized emulsion is subjected to drying, providing the curcumin formulation in a dry form, preferably powder. Drying may be carried out by a number of methods known in the state of the art, but preferably spray drying is carried out using temperature at the inlet of the dryer in the range of 110 - 160 °C, preferably 150 °C, and not more than 65 °C at the outlet, preferably 55 °C.

The invention also provides a curcumin formulation in a dry form obtained by the method of production of the curcumin formulation according to the invention.

The formulation obtained as described above is based solely on ingredients approved for use in the food industry in accordance with the EU law. At the same time, due to the use of only plant ingredients and the complete elimination of the use of ethyl alcohol as a solvent in the formulation production process, the formulation can be certified according to the HALAL procedures, what in the case of its use in food products intended for Muslim countries markets, enables their distribution due to non-use of ethyl alcohol as a solvent in the formulation production process. The formulation is characterized by an intense orange-yellow color, natural for turmeric.

Unlike the known turmeric formulations available on the market, the turmeric formulation according to the invention, the food products based on it, as well as the emulsion produced from it, is characterized by a completely neutral taste – without the bitterness characteristic of turmeric extracts and without the sweetish aftertaste, typical for formulations with maltodextrin and/or maltose, what significantly increases the scope of its use in the food and pharmaceutical industries, and dietary supplement production. The formulation shows stability in a dry form and in the form of an aqueous solution, which enables its effective use in a wide range of food products.

The invention relates to a beverage concentrate comprising the curcumin formulation according to the invention.

The preferred beverage concentrate further comprises at least one of the following:

- a stabilizer and thicker in the concentration of 0-28% w/w, preferably in the concentration of 0-28% w/w, preferably glycerol and/or xanthan gum;
- an antioxidant and acidity regulator 0-5% w/w, preferably ascorbic acid;
- an acidity regulator 0-5% w/w, preferably citric acid;
- an electrolyte and sweetener 0-80% w/w, preferably glucose;
- B vitamins, preferably selected from thiamine, niacin, pyridoxine, folic acid and riboflavin;
- preservatives, preferably benzoic acid, sodium benzoate, potassium sorbate, or a mixture thereof.

The invention also relates to a beverage comprising the curcumin formulation according to the invention or obtained by dilution of a beverage concentrate according to the invention.

The preferred beverage is a beverage based on fruit juices, water, a milk beverage, a plant beverage substituting milk.

The preferred beverage is a functional beverage supporting liver function, wherein preferably, the concentration of curcuminoids in the beverage is 0.01-0.1% w/w.

The invention relates to a dietary supplement comprising the curcumin formulation according to the invention, preferably in a liquid form, preferably the supplement is consumed in a dose corresponding to 0.2 to 2.0 g of curcuminoids per kilogram of body weight of the individual.

The preferred dietary supplement supports liver function, wherein preferably, it activates the action of liver enzymes, preferably the enzymes of the dehydrogenases group, preferably aldehyde dehydrogenase.

The preferred dietary supplement supports the anti-inflammatory effect and is intended for anti-inflammatory prevention and reduction of inflammation including the prevention of complications and severe course of viral infections, including coronavirus infections, including SARS-CoV-2, preferably is for use in the period of increased viral infections incidence, preferably in the form of capsules comprising curcumin formulation according to the invention in the dry form.

The preferred dietary supplement is in liquid form and contains the formulation according to the invention in an amount that provides 0.01-0.1% w/v of curcuminoids, fruit juice concentrate 1-20% w/v, sweetener 2-10% w/v, stabilizing and thickening agent in a concentration of 2-10% w/v.

The invention also relates to a pharmaceutical composition comprising the curcumin formulation according to the invention and a pharmaceutically acceptable additive.

The invention also relates to pharmaceutical composition comprising the curcumin formulation according to the invention for use as a medicine.

Preferably, the composition is for use as a medicine in the regulation of liver function, a medicine with an anti-inflammatory, anticancer and antidepressant effect.

Preferably, the composition is for use as a medicine in the regulation of liver function, activating the action of liver enzymes, preferably the enzymes of the dehydrogenases group, preferably aldehyde dehydrogenase.

The invention also relates to a pharmaceutical composition comprising the curcumin formulation according to the invention for use as a liver function stabilizer, an agent supporting the liver, an agent activating the action of liver enzymes, preferably the enzymes of the dehydrogenases group, preferably aldehyde dehydrogenase.

The invention also relates to a pharmaceutical composition comprising the curcumin formulation according to the invention for use as agent supporting the anti-inflammatory effect intended for anti-inflammatory prevention and reduction of inflammation, preferably the prevention of complications and severe course of viral infections, including coronavirus infections, including SARS-CoV-2, wherein it is preferably used in the period of increased viral infections incidence.

The invention also relates to a dye comprising the curcumin formulation according to the invention.

The preferred dye is a food dye used in medicine, cosmetics, food, animal feed.

The invention also relates to the use of the curcumin formulation according to the invention as a functional additive to a beverage, as a dye, as an active ingredient in a dietary supplement, as an active ingredient in a pharmaceutical composition, as a food additive, as a feed additive.

The invention also relates to the use of the curcumin formulation according to the as an anti-hangover agent, an agent for prevention and/or alleviation -hangover symptoms or as an agent activating liver enzymes, preferably the enzymes of the dehydrogenases group, preferably aldehyde dehydrogenase.

What is extremely important due to the increased bioavailability, beverages/emulsions or compositions produced based on the turmeric formulation according to the invention, do not require the addition of piperine, what was necessary to increase the bioavailability of turmeric formulations known in the state of the art.

The inventors have found that the formulation according to the invention can be successfully used to produce a stable concentrate comprising from 5 to 20% of the formulation concentration and the final concentration of curcuminoids (from 0.1 to 2.0 % w/w) constituting, preferably, a 5 to 20 times concentrated concentrate for use in the production of beverages, especially functional beverages, in which curcumin is the active substance, and especially beverages with an activating effect on acetaldehyde metabolism.

Due to the high stability, no modification of the biological activity and increased bioavailability of curcuminoids, the formulation according to the invention can be used to produce functional dietary supplements or pharmaceutical preparations, in which curcumin is one of the active substances. Preferably, such preparations include preparations regulating liver function, activating the action of liver enzymes, including enzymes of the dehydrogenases group, including aldehyde dehydrogenase, with an anti-inflammatory, anticancer and antidepressant effect.

In the preferred embodiment, the formulation is used to produce a functional beverage in the form of a dietary supplement supporting liver function, particularly in the field of activation of acetaldehyde-degrading enzymes, preventing some of the negative aspects of alcohol consumption, by lowering the concentration of acetaldehyde in the blood. In the preferred embodiment, the formulation is used in a concentration corresponding to the final concentration of curcuminoids at the level of 0.01-0.1% w/w.

Due to the increased bioavailability of curcumin in the form of the described formulation, the use of the formulation enables elimination from the recipe of the functional beverage concentrate and the functional beverage according to the invention, of substances supporting the bioavailability, used in the preparations available on the market, in the form of, e.g., black pepper extract in the form of piperine (ReadyQ, alcorythm® preparations) adversely affecting the taste of the preparation due to its sharp and burning taste, and acting as an irritant, as well as the substances additionally inducing the activity of enzymes accelerating the decomposition of acetaldehyde, such as DHM (dihydromyricetin) (ReadyQ, Morning Recovery preparations).

The concentrate may also contain excipients in any combination, comprising at least one of the following substances, such as:

- a stabilizer and thicker in the concentration of 0-20, preferably 0--28% w/w, preferably glycerol and/or xanthan gum,

- an antioxidant and acidity regulator 0-5% w/w, preferably ascorbic acid,

- an acidity regulator 0-5% w/w, preferably citric acid,

- an electrolyte and sweetener 0-80% w/w, preferably glucose.

The concentrate may also contain B vitamins (including thiamine, niacin, pyridoxine, folic acid and riboflavin) in the concentrations for each of the vitamins of not more than twice the maximum reference dietary intake, taking into account the excess stability multiplied by the value of the concentrate concentration.

The concentrate may contain preservatives in any combination, comprising at least one of the following substances, assuming that the total concentration of preservatives, after diluting the concentrate to the target concentration will not exceed 0.2% w/w, the concentrate may comprise such preservatives as: benzoic acid in the concentration of 0-2% w/w, sodium benzoate in the concentration of 0-2% w/w, potassium sorbate in the concentration of 0-2% w/w, or a mixture thereof.

The use of the formulation according to of the invention as a yellow dye in the food, pharmaceutical and cosmetic industries for dyeing food and pharmaceutical and cosmetic products based on water turned out to be particularly preferable. Due to the completely neutral taste, the use of excipients allowed in the food industry, the natural color of the formulation according to the invention resulting from the lack of blockage of chromophore groups, and the high stability of the aqueous solution of the formulation according to the invention, the aqueous solution of the formulation according to the invention can be used at the target concentration of *quantum satis*, for example as a dye E100 in the production of beverages (including beverages based on natural or reconstituted fruit juices with a pH in the range of 3.0-7.0), for dyeing cakes, pastries, dyeing tablet mass and syrups as well as cosmetics (e.g., shampoos, creams, masks).

"Golden milk" – that is a product obtained from the water extract of turmeric rhizome in milk (ruminant milk), or a plant milk substitute (a beverage obtained based on the cereal grains, nuts, legume seeds) is used, in the unconventional medicine and dietetics based on the medical traditions of the Far East, as a beverage that increases the body's immunity and counteracts

inflammation. The bitter-burning taste of turmeric, characterizing the beverage, discourages the use of the preparation as a preventive and therapeutic agent. An additional disadvantage of the beverage is the relatively low content and low bioavailability of curcuminoids in the water extract from the turmeric rhizome.

Due to the completely neutral taste, the use of excipients allowed in the food industry and bioavailability, the inventors have found that the use of the formulation according to the invention to produce beverages, and especially dairy beverages, allows to overcome the technological problem arising from the state of the art. The formulation according to the invention forms stable suspensions in animal milk (preferably ruminant milk, particularly preferably in cow's, buffalo's, sheep's or goat's milk) and in plant milk substitutes in the form of beverages obtained from cereal grains, nuts, legume seeds (preferably oat, proso millet, hemp, cashews, pecans, almonds, soybeans), including beverages containing additives and excipients, as well as supplements in the form of calcium salts, thickeners, stabilizers and sweeteners In the preferred embodiment, the milk is cow's milk, almond, soy or oat beverage, and the concentration of the formulation in the final product is 0.01%-1.0%, particularly preferable 0.1%. Following dissolution of the formulation, the product is subjected to the pasteurization process

Due to the completely neutral taste, excellent physical and physiological properties including excellent bioavailability, formulation of curcumin in the form of the described formulation according to the invention may be use as excipients allowed in the food industry, dye industry, feed additive, nutraceuticals, functional food, medicament for example for regulation of liver function, a medicine with an anti-inflammatory, anticancer and antidepressant and may other uses the curcumin is used. The inventors have found that the formulation according to the invention avoids the technological problems resulting from the state of the art and work perfectly in all above indicated uses and examples.

DESCRIPTION OF FIGURES

For a better understanding of the invention, it has been illustrated in the nonlimiting embodiments and in the accompanying figures, in which:

FIG. 1 shows the particle size distribution of the produced emulsion, prior to spray drying;

FIG. 2 shows the particle size distribution in the dissolved powder comprising 15% of curcuminoids in its composition.

EXAMPLES

The following examples were presented merely to illustrate the invention and to clarify its various aspects, but are not intended to be limitative, and should not be equated with all its scope, which is defined in the appended claims. In the following examples, unless it was otherwise indicated, standard materials and methods used in the field were used, or the manufacturers' instructions for specific materials and methods were followed.

Example 1

Production of the curcumin formulation in powder form

6 g of starch octenylsuccinate (e.g., PURITY GUM 2000®) and 2 g of hydroxypropyl methylcellulose (a cellulose derivative with the viscosity in the range of 40 – 200 mPas). After mechanical mixing of the polymers, they were dissolved in a polar solvent, which was water, in the amount of 100 ml. The obtained mixture was mechanically mixed to dissolve and combine polymers. The inventors have unexpectedly found, that this occurs after not less than 10 hours, preferably 12 hours. 1.6 g of turmeric extract (min. 90% of curcuminoids content) was dissolved in 30 ml of ethyl acetate at the boiling temperature of this solvent. The dissolved curcuminoids extract was added immediately after dissolution to the mixture of polymers and homogenized using a mechanical homogenizer at the rotational speed of the working element of 10 000 rpm to the moment of formation of a pre-emulsion, that is obtaining a homogeneous mixture evaluated by sight. The entire mixture, from the moment the curcuminoids were introduced into the mixture of polymers, was cooled in an ice water bath. The temperature was kept in the range of 5-10 °C, preferably about 7 °C. After the production of the pre-emulsion, it was subjected to proper homogenization using a turbo-vortex homogenizer and the process parameters: 22 000 rpm for 3 minutes, after which time water or 1.5% of an aqueous solution of β-glucan from oat or barley with a low viscosity (40 – 200 mPas) was added in an amount of 50 ml and further homogenized for 1 min. The emulsion prepared in this way was immediately combined with a 1% aqueous

solution of carboxymethyl cellulose (CMC) with a high viscosity (> 5600 mPas). The emulsion was diluted in 1 l of the solution prepared in this way by mechanical stirring using a propeller stirrer. For the emulsion solution, the percentage distribution of particle sizes in the sample in the particular size categories was determined. Results with a distribution similar to the results from the example 2 were obtained (over 99% of the particles had a size in the range from about 0.1 to about 1 µm).

The emulsion solution prepared in this way was subjected to classical spray drying, using the temperature of about 150 °C at the inlet of the dryer, and about 55 °C at the outlet.

For the obtained curcumin formulations in powder form, after its dissolution, the size of the obtained particles and their size distribution were determined (**Table 1, Fig 2**).

Table 1. Percentage distribution of particle sizes in the sample (powder after dissolution), in the particular size categories.

No.	Lower limit of the range (µm)	Average particle size in the range (µm)	% of particles	No.	Lower limit of the range (µm)	Average particle size in the range (µm)	% of particles	No.	Lower limit of the range (µm)	Average particle size in the range (µm)	% of particles
1	0	0.025	0	23	1.24	1.34	0	45	35.56	38.495	0
2	0.05	0.055	0	24	1.44	1.56	0.07	46	41.43	44.85	0
3	0.06	0.065	0	25	1.68	1.815	0.31	47	48.27	52.25	0
4	0.07	0.075	0	26	1.95	2.115	0.43	48	56.23	60.87	0
5	0.08	0.085	0	27	2.28	2.465	0.43	49	65.51	70.915	0
6	0.09	0.1	0	28	2.65	2.87	0.36	50	76.32	82.615	0
7	0.11	0.12	0	29	3.09	3.345	0.28	51	88.91	96.245	0
8	0.13	0.14	0.01	30	3.6	3.895	0.2	52	103.58	112.125	0
9	0.15	0.16	0.07	31	4.19	4.535	0.13	53	120.67	130.625	0
10	0.17	0.185	0.51	32	4.88	5.285	0	54	140.58	152.175	0
11	0.2	0.215	2.39	33	5.69	6.16	0	55	163.77	177.285	0
12	0.23	0.25	7.71	34	6.63	7.175	0	56	190.8	206.54	0
13	0.27	0.29	15.99	35	7.72	8.36	0	57	222.28	240.615	0
14	0.31	0.335	21.02	36	9	9.74	0	58	258.95	280.315	0
15	0.36	0.39	19.72	37	10.48	11.345	0	59	301.68	326.57	0
16	0.42	0.455	15.52	38	12.21	13.215	0	60	351.46	380.455	0
17	0.49	0.535	9.77	39	14.22	15.395	0	61	409.45	443.23	0
18	0.58	0.625	4.02	40	16.57	17.94	0	62	477.01	516.36	0
19	0.67	0.725	1	41	19.31	20.9	0	63	555.71	601.56	0
20	0.78	0.845	0	42	22.49	24.345	0	64	647.41	700.82	0
21	0.91	0.985	0	43	26.2	28.365	0	65	754.23	816.45	0
22	1.06	1.15	0	44	30.53	33.045	0	66	878.67	878.67	0

It was unexpectedly found, that the formulation obtained according to the above description is characterized by a particularly preferred distribution of particle sizes in the diameter range from about 0.1 to about 1 µm (over about 95% of the particles, preferably over 97%, preferably 100%

of the particles), with the curcuminoids content in the range (from 0.01% to 15%), what increases its dispersion properties in an aqueous environment, enabling the obtaining of an emulsion with a high degree of homogeneity.

The quantitative and qualitative composition of the obtained, dried (up to 5% humidity) formulation produced according to the above example was as follows:

starch octenylsuccinate 29.5% w/w

hydroxypropyl methylcellulose (as a cellulose derivative) 9.8% w/w

turmeric extract (min. 90% of curcuminoids) 7.9% w/w

β-glucan 3.7% w/w

carboxymethyl cellulose 49.1% w/w;

wherein over 97% of the particles after dissolution had a size in the range from about 0.1 to about 1 μm.

The formulation was characterized by an intense orange-yellow color, was neutral in taste and easily dissolved in water forming a homogenous suspension.

Example 2

Production of the curcumin formulation in powder form

The formulation of the mixture was obtained as in example 1, with the difference that:

- after mechanical mixing of the polymers, they were dissolved in a polar solvent, which was water, in the amount of 90 ml;
- 2 g of turmeric extract (min. 90% of curcuminoids content) was dissolved in a mixture of ethyl acetate and acetone in the 5:1 ratio, in the amount of 35 ml at the boiling temperature of the mixture.

The obtained mixture was mechanically mixed to dissolve and combine polymers, for not less than 10 hours, preferably 12 hours. The dissolved curcuminoids extract was added immediately after dissolution to the mixture of polymers and homogenized using a mechanical homogenizer at the rotational speed of the working element of 10 000 rpm to the moment of formation of a pre-emulsion, that is obtaining a homogeneous mixture evaluated by sight. The entire mixture, from the moment the curcuminoids were introduced into the mixture of polymers, was cooled

in an ice water bath. The temperature was kept in the range of 5-10 °C, preferably about 7 °C. For the obtained curcumin formulations in the form of emulsion, the size of the obtained particles and their size distribution were determined (**Table 2**, **Fig 1**). The inventors have unexpectedly found that the formulation obtained according to the above description is characterized by a particularly preferred distribution of particle sizes in the diameter range from about 0.1 to about 1 µm (over about 95% of the particles, preferably 100% of the particles), with the curcuminoids content in the range (from 0.01% to 20%), what increases its dispersion properties in an aqueous environment, enabling the obtaining of an emulsion with a high degree of homogeneity.

Table 2. Percentage distribution of particle sizes in the sample (emulsion), in the particular size categories.

No.	*Lower limit of the range (µm)*	*Average particle size in the range (µm)*	*% of particles*	*No.*	*Lower limit of the range (µm)*	*Average particle size in the range (µm)*	*% of particles*	*No.*	*Lower limit of the range (µm)*	*Average particle size in the range (µm)*	*% of particles*
1	0	0.025	0	23	1.24	1.34	0	45	35.56	38.495	0
2	0.05	0.055	0	24	1.44	1.56	0	46	41.43	44.85	0
3	0.06	0.065	0	25	1.68	1.815	0	47	48.27	52.25	0
4	0.07	0.075	0	26	1.95	2.115	0	48	56.23	60.87	0
5	0.08	0.085	0	27	2.28	2.465	0	49	65.51	70.915	0
6	0.09	0.1	0	28	2.65	2.87	0	50	76.32	82.615	0
7	0.11	0.12	0	29	3.09	3.345	0	51	88.91	96.245	0
8	0.13	0.14	0.02	30	3.6	3.895	0	52	103.58	112.125	0
9	0.15	0.16	0.11	31	4.19	4.535	0	53	120.67	130.625	0
10	0.17	0.185	0.57	32	4.88	5.285	0	54	140.58	152.175	0
11	0.2	0.215	2.26	33	5.69	6.16	0	55	163.77	177.285	0
12	0.23	0.25	6.53	34	6.63	7.175	0	56	190.8	206.54	0
13	0.27	0.29	12.97	35	7.72	8.36	0	57	222.28	240.615	0
14	0.31	0.335	17.42	36	9	9.74	0	58	258.95	280.315	0
15	0.36	0.39	17.9	37	10.48	11.345	0	59	301.68	326.57	0
16	0.42	0.455	16.65	38	12.21	13.215	0	60	351.46	380.455	0
17	0.49	0.535	13.67	39	14.22	15.395	0	61	409.45	443.23	0
18	0.58	0.625	8.53	40	16.57	17.94	0	62	477.01	516.36	0
19	0.67	0.725	3.38	41	19.31	20.9	0	63	555.71	601.56	0
20	0.78	0.845	0	42	22.49	24.345	0	64	647.41	700.82	0
21	0.91	0.985	0	43	26.2	28.365	0	65	754.23	816.45	0
22	1.06	1.15	0	44	30.53	33.045	0	66	878.67	878.67	0

The quantitative and qualitative composition of the obtained, dried (up to 5% humidity) formulation produced according to the above example was as follows:

starch octenylsuccinate 28.9% w/w;

hydroxypropyl methylcellulose 9.6% w/w;

turmeric extract (min. 90% of curcuminoids) 9.6% w/w;

β-glucan 3.6% w/w;

carboxymethyl cellulose 48.3% w/w;

wherein in this composition in the form of an emulsion, 100% of the particles had a size in the range from about 0.1 to about 1 µm.

For the obtained curcumin formulation in powder form, the size of the obtained particles and their size distribution after dissolution were determined. Over 98% of the particles after dissolution of the powder had a size in the range from about 0.1 to about 1 µm. Therefore, results with a distribution similar to the results from the **example 1** were obtained.

The formulation was characterized by an intense orange-yellow color, was neutral in taste and easily dissolved in water forming a homogenous suspension.

Example 3

Production of the curcumin formulation in powder form

The formulation of the mixture was obtained as in **example 1**, with the difference that:

- after mechanical mixing of the polymers, they were dissolved in a polar solvent, which was water, in the amount of 90 ml;
- the prepared emulsion solution was subjected to freeze drying using the following process parameters: drying pressure 0.5 mbar, shelf temperature program: from 0 to 2 h -20 °C, 2-3 h 20 °C, 3-18 h 20°C, 18-19 h 40 °C, 18-19 h pressure 0.2 mbar, 19-24 h 40 °C – end of the process. For the obtained curcumin formulations in the form of emulsion and powder, after dissolution of the powder, the size of the obtained particles and their size distribution were determined. Results with a distribution similar to the results in examples 1 and 2 were obtained.

The inventors have unexpectedly found that the formulation obtained according to the above description is characterized by a particularly preferred distribution of particle sizes in the diameter range from about 0.1 to about 1 µm (over about 95% of the particles, preferably 100% of the particles), with the curcuminoids content in the range (from 0.01% to 20%), what increases its dispersion properties in an aqueous environment, enabling the obtaining of an emulsion with a high degree of homogeneity.

The quantitative and qualitative composition of the obtained, dried (up to 5% humidity) formulation produced according to the above example was as follows:

starch octenylsuccinate 26.5% w/w

hydroxypropyl methylcellulose 9.8% w/w

turmeric extract (min. 90% of curcuminoids) 16.4% w/w

β-glucan 3.7% w/w

carboxymethyl cellulose 43.6% w/w

wherein after dissolution over 97% of the particles had a size in the range from about 0.1 to about 1 µm.

The formulation was characterized by an intense orange-yellow color, was neutral in taste and easily dissolved in water forming a homogenous suspension.

Example 4

Aqueous solution of the curcumin formulation

Curcumin formulations in powder form, prepared according to examples 1-3, were dissolved in water in certain concentrations and their physicochemical properties were analyzed. Based on the obtained results, it was found that the formulations form a stable suspension in an aqueous environment, in the concentration range from 0.01% to 30% w/w. Preferably, the concentration is 0.05-10.00%, particularly preferably 0.5%-1.5% w/w. In the indicated concentrations, particularly in the preferred concentrations, the formulation forms an optically homogeneous suspension, at room temperature after 24 hours, no visible precipitate formation was observed. The aqueous solution of the formulation in the preferred concentrations is characterized by chemical stability (the study was conducted based on the observation of color changes, indicating the oxidation processes of the active substance and color changes in an acidic environment, in the pH range equal to or higher than 3.0 and lower than 7.0, what enables its use in food production.

In aqueous solutions, the formulation is stable for a period of min. 12 months, as shown in aging tests, under accelerated aging conditions (according to the guideline ICH Q1A (R2) Stability testing of new drug substances and drug products – EDQM recommendations).

In order to confirm the stability of the suspension and to determine the level of sedimentation, a test, consisting of centrifugation of the suspension with the curcuminoids concentration of 0.5% in 15 ml of distilled water, at the speed of 5000 rpm, for 20 min at the temperature of 25 °C, in the REF 11778 rotor of the MPW380R centrifuge, manufactured by MPW, was carried out.

The following suspensions were compared:

a) of the tested formulation from **examples 1-3**
b) of the turmeric extract comprising 95% of curcuminoids
c) of the CAVACURMIN preparation (Wacker Chemie AG) and the CurQfen preparation (Akay Group Ltd.)

The suspensions were standardized to the total curcuminoids concentration of 0.5% w/w.

Table 3. Stability results for the developed formulation and similar formulations available on the market. P1, P2, P3 - formulation produced according to the example 1, 2, 3, respectively.

Tested product	Sample concentration (volume of sample subjected to the stability test 15 ml)	Concentration of curcuminoids in an aqueous suspension (15 ml)	% of substance in the form of a sediment after centrifugation at 50000 rpm at 25 °C and after 20 min
Formulation according to the invention			
P1 (7.9% of curcuminoids*)	949.33 mg in 15 ml of water	0.5%	**20.4%**
P2 (9.6% of curcuminoids*)	781.22 mg in 15 ml of water	0.5%	**19.9%**
P3 (16.4% of curcuminoids*)	457.3 mg in 15 ml of water	0.5%	**19.5 %**
CAVACURMIN® (16.4% curcuminoids *)	457.3 mg in 15 ml of water	0.5 %	100 %
CurQfen® (47.1% curcuminoids *)	159.2 mg in 15 ml of water	0.5%	95 %
turmeric extract (%curcuminoids *)	81.5 mg in 15 ml of water	0.5 %	The extract floats on the surface of water

*Curcuminoid content was measured spectrophotometrically at λ = 420 nm. The solution used for curcuminoids extraction from the formulation, and in which the spectrophotometric measurements were made was: water:acetone 30:70 v/v.

The results unambiguously indicate a greater stability of the suspension of the formulation according to the invention.

Example 5

Bioavailability of the aqueous suspension of the formulation

The inventors conducted a bioavailability study of the suspension of the formulation in water according to the invention. The test was performed by *in vitro* method on a cellular model of absorption and permeation through the layer of epithelial cells of intestinal epithelium, using the Caco-2 cell line, according to the protocol described in: Frank, J., 2017. In the study, the bioavailability of a turmeric extract, the tested formulation according to the invention, and the preparation CAVACURMIN® (Wacker Chemie AG) and CurQfen® (Akay Group Ltd.) was compared. The bioavailability was tested on suspensions normalized to the final curcuminoids content – the concentration of 10 µg/ml.

Table 4. Absorption and permeability of the curcuminoids formulations through a monolayer of the Caco-2 cells over 60 min. P1, P2, P3 - formulation produced according to the example 1, 2, 3, respectively.

	Initial concentration of curcuminoids	**Time (min)**	**% of curcumin absorbed by the cells**	**% of curcumin, which has penetrated through a monolayer of cells**
CurQufen®	10 µg/ml	60	39.64 % ± 0.22%	0.66 % ± 0.04%
CAVACURMIN®	10 µg/ml	60	21.61 % ± 0.15%	0.00 % ± 0.00 %
Formulation according to the invention **P1** **P2** **P3**	10 µg/ml	60	 42.57 % ± 1.21% 44.23 % ± 0.89% 43.62 % ± 0.07%	 0.63% ± 0.80% 0.72 % ± 0.40% 0.67 % ± 0.50%

The study unambiguously showed an increased bioavailability of the formulation according to the invention compared to the CAVACURMIN® preparation and the CurQufen® preparation. The formulation according to the invention is absorbed by the cells at a level over 2 times higher than the CAVACURMIN® preparation and more than 10% higher than the CurQfen® preparation (it

was shown in the studies, that the CurQfen® preparation has a 45 to 270 times higher bioavailability than the native turmeric extract – Kumar et al., 2016). At the same time, the formulation according to the invention shows a 1.5% higher permeability through a cellular monolayer than the CurQfen® preparation, measured over 60 min.

Example 6

A concentrate for use in the production of functional beverages and dietary supplements using the formulation produced according to one of the methods described in the examples 1, 2 and 3.

A 5-fold concentrated concentrate was prepared, comprising:

the formulation according to the invention : 5% w/w produced according to the **Example 1**:

xanthan gum: 1.1% w/w;

glycerol: 26.5% w/w;

citric acid: 2.5% w/w;

sodium benzoate: 0.5% w/w;

potassium sorbate: 0.5% w/w;

glucose: 40% w/w;

water: up to 100% w/w.

The concentrate showed a high stability, confirmed by pH and microbiological tests. No changes in the pH or growth of mesophilic bacteria, molds and yeasts were found during storage up to 6 months. The color of the concentrate and its other organoleptic properties also did not change. In order to prepare the final beverage, the concentrate should be diluted in the proportion 1:4 (concentrate : liquid), preferably with water or reconstituted fruit juice from concentrate, preferably tropical fruit juice, preferably mango, peach or orange juice, or a mixture of these juices. Similarly, concentrates were prepared using the formulation prepared according to Example 2 (using 7% w/w of the prepared formulation) and Example 3 (using 10% w/w of the prepared formulation).

Example 7

The use of the formulation according to the invention as a dye

The formulation according to the invention was used as a food dye E-100 (curcumin) in accordance with the concentrations allowed in the food law, i.e. 100 mg/l(kg) in beverages based on juices, 150 mg/l(kg) in products based on milk.

a) 5-fold concentrated fruit juice concentrate (preferably orange, peach, pineapple, particularly preferably orange) was mixed with the formulation produced according to the **example 1,** so that the final concentration of the formulation in the final product was 0.01% w/v, the mixture was made up with water to obtain a 1-fold final concentration of the juice.

b) The formulation according to the invention was added to a fruit yogurt (preferably peach or mango flavored) to obtain the final concentration of 0.015% w/w. The obtained mixture was subjected to stirring and homogenization.

Example 8

The use of the formulation according to the invention for the production of a beverage with curcumin based on milk or a plant beverage substituting milk.

a) 10-fold concentrated cow's milk was mixed with the formulation produced according to the **example 1**, so that the final concentration of the formulation in the final product was 0.015%. The mixture was made up with water to obtain a 1-fold final concentration of the milk.

10-fold concentrated plant beverage concentrate substituting milk (preferably a soybean, almond or oat beverage) was mixed with the formulation produced according to the **example 1,** so that the final concentration of the formulation in the final product was 0.01%. The mixture was made up with water to obtain a 1-fold final concentration of the beverage

Example 9

Functional beverage

A functional beverage supporting liver function was produced, activating the action of liver enzymes, preferably the enzymes of the dehydrogenases group, preferably aldehyde dehydrogenase, before or during alcohol consumption. a) 5-fold concentrated fruit juice

concentrate (preferably lemon, grapefruit, mango, passion fruit, orange, peach, pineapple, particularly preferably orange) was mixed with the formulation produced according to the **example 1,** so that the final concentration of the formulation in the final product was 0.01% w/v, the mixture was made up with water to obtain a 1-fold final concentration of the juice.

b) 5-fold concentrated fruit juice concentrate (preferably lemon, grapefruit, mango, passion fruit, orange, peach, pineapple) was mixed with the formulation prepared according to **example 1** so that the final formulation concentration in the final product was 0.01% w/v, the mixture was made up with water so as to obtain a 1-fold final concentration of juice, then to the mixture a sweetener (glucose) to a concentration of 4% in the final product and citric acid to a concentration of 0.4% in the final product were added, and then made up to obtain a double-diluted fruit juice.

The functional juice-like beverage can be consumed without limitation as a fruit juice equivalent to dilute alcohol-based cocktails or to sip alcohol, in order to activate the metabolism of acetaldehyde, preferably the enzymes of the dehydrogenases group, preferably aldehyde dehydrogenase, and to reduce the effects of alcohol consumption resulting from acetaldehyde poisoning.

Example 10

Dietary supplement in the form of capsules

A dietary supplement in the form of capsules was developed to support the liver, activating the action of liver enzymes, including enzymes of the dehydrogenases group, especially aldehyde dehydrogenase, intended to be taken before or during alcohol consumption. The formulation prepared according to **example 1** was encapsulated in pharmaceutically acceptable gel capsules in doses of 500 mg of the substance according to the formulation. In an optional variant, the gel capsules contain, apart from the formulation, the addition of adjuvants and fillers, preferably the adjuvant is beta-glucan in a dose of up to 100 mg, and the filler is cellulose. Preferably, the capsules are made of a gel material that dissolves within the small intestine, preferably the substance is a polymer of plant, microbial or synthetic origin meeting the HALAL and vegan certification criteria. A single gel capsule is a single dose of the supplement to be taken during

the period from 60 minutes before the start of alcohol consumption or during its consumption for an adult with a body weight of up to 85 kg. Taking the capsules, especially before the consumption of alcohol, activates the action of liver enzymes, including enzymes of the dehydrogenases group, increasing the tolerance of a person taking a curcumin formulation with increased bioavailability to ethyl alcohol consumed by the individual.

Example 11

Dietary supplement in liquid form

A liquid dietary supplement was prepared to support liver function, to activate the action of liver enzymes, preferably enzymes from the dehydrogenase group, preferably aldehyde dehydrogenase, to be taken before or during alcohol consumption.

Mango juice concentrate was mixed in a container with a vortex mixer with water, glucose, glycerol, xanthan gum, citric acid, B vitamins, thiamine, niacin, pyridoxine, folic acid and riboflavin, sodium benzoate preservatives and potassium sorbate, as well as the formulation according to **example 1** to the final concentration curcuminoids of 0.1% w/v, and the natural passion fruit flavor. After thorough mixing of the ingredients, the prepared liquid dietary supplement with a density of 1.02 $g*cm^{3\ -1}$ was poured into 100 ml bottles, which constitute a single dose of the supplement to be taken from 60 minutes before the start of alcohol consumption or during its consumption for an adult with a body weight of up to 85 kg.

The final concentrations of ingredients in the ready liquid dietary supplement were, respectively:

- mango juice concentrate 6% w/v
- natural passion fruit flavor 0.001% w/v
- glucose 8% w/v
- xanthan gum 0.226% w/v
- glycerol (99.5%) 5.333% w/v
- citric acid 0.5% w/v
- potassium sorbate 0.1% w/v
- sodium benzoate 0.1% w/v
- niacin 0.0165% w/v

- thiamine 0.00296% w/v
- riboflavin 0.00132% w/v
- pyridoxine 0.023%w/v
- folic acid 0.000352% w/v
- formulation according to the invention according to example 1 to a final concentration of curcuminoids of 0.1% w/v

The dietary supplement in a liquid form is intended for consumption immediately before or during alcohol consumption in the form of a single 100 ml dose for an adult with a body weight of up to 85 kg, as a preparation supporting the liver, activating the action of liver enzymes, preferably the enzymes of the dehydrogenases group, preferably dehydrogenase aldehyde for use before or during alcohol consumption. Taking the dietary supplement in a liquid form, especially before alcohol consumption, activates the action of liver enzymes, including enzymes of the dehydrogenases group, increasing the tolerance of a person taking the curcumin formulation according to the invention, with increased bioavailability, ,contained therein, to ethyl alcohol consumed by the individual.

Example 12

Dietary supplement in the form of capsules- as anti-inflammatory agent

A dietary supplement in the form of capsules with an anti-inflammatory effect was produced, intended for anti-inflammatory prophylaxis, including the prevention of complications and severe course of viral infections, including coronavirus infections, including SARS-CoV-2 virus, to be taken in the period of increased viral infections incidence.

The formulation prepared according to **example 1** was encapsulated in pharmaceutically acceptable water-soluble biopolymers (gelatin gel capsules, hypromellose hard capsules containing powder) in doses of 450 mg capsules according to the formulation.

Preferably, the capsules are made of a gel material that dissolves within the small intestine, preferably the substance is a polymer of plant, microbial or synthetic origin meeting the HALAL and vegan certification criteria. A single gel capsule is a single dose of the supplement to be taken daily in the period of increased incidence of viral infections. Taking the capsules, especially by

people belonging to the group of increased risk of developing the disease, especially people who have confirmed at least one of the disease risks, such as hypertension, obesity, type 2 diabetes, hypercholesterolaemia, leads to a reduction of inflammation from the above. diseases, and thus the disease with the virus of post-infectious complications and the severe course of infection, in the course of viral infections, especially coronavirus infections, especially infection with the SARS-CoV-2 virus.

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Claims:

1. A curcumin formulation in a dry form, characterized in that it comprises

- starch octenylsuccinate in the amount of 0.1-45% w/w,

-cellulose derivative 0.01-20% w/w in the form of hydroxypropyl methylcellulose, hydroxypropylcellulose or a mixture thereof;

- turmeric extract in the amount of 0.01-20% w/w, wherein the turmeric extract comprises at least 90% w/w of curcuminoids, including curcumin, demethoxycurcumin and bisdemethoxycurcumin,

- β-glucan in the amount of 0.1-10% w/w,

- cellulose derivative with the viscosity of >5600 mPas in the amount of 10-70% w/w, selected from carboxymethyl cellulose, methylcellulose, ethylcellulose, hydroxypropylcellulose, ethyl methyl cellulose, cross-linked cellulose gum, preferably, the cellulose derivative with the viscosity of >5600 mPas is carboxymethyl cellulose;

wherein at least 90%, more preferably at least 95% of the particles, preferably at least 99% of the particles have a diameter below 1 μm.

2. The curcumin formulation in a dry form according to claim 1, characterized in that, it comprises

- starch octenylsuccinate in the amount of 25-35% w/w, more preferably 28-32% w/w;

- cellulose derivative in the form of hydroxypropyl methylcellulose, hydroxypropylcellulose or the mixture thereof, in the amount of 5-15% w/w, preferably in the amount of 8-10% w/w;

- turmeric extract in the amount of 5-15% w/w, preferably in the amount of 8-10% w/w, wherein the turmeric extract comprises at least 90% w/w of curcuminoids, including curcumin, demethoxycurcumin and bisdemethoxycurcumin,

- β-glucan in the amount of 3-5% w/w, preferably in the amount of 3.8-4.2% w/w;

- cellulose derivative with the viscosity of over 5600 mPas in the amount of 30-60% w/w, preferably in the amount of 48-52% w/w, preferably, the cellulose derivative with the viscosity of >5600 mPas is carboxymethyl cellulose.

3. The curcumin formulation in a dry form according to claims 1-2, characterized in that, in the curcumin formulation in the dry form at least 90%, more preferably at least 95% of the particles, preferably at least 99% of the particles have the diameter in the range from 0.1 to about 1 μm.

4. A method of production of a curcumin formulation in a dry form, which comprises the following steps:

a) turmeric extract with a minimum content of 90% of curcuminoids is dissolved in an organic solvent, approved for use in food production;

wherein the preferred organic solvent used to dissolve the turmeric extract in step a) is ethyl acetate, methyl acetate, butyl acetate, ethanol, butanol, more preferably ethyl acetate;

b) the dissolved turmeric extract from step a) is added to a solution of a carrier, which is a mixture of polymers comprising starch octenylsuccinate and the cellulose derivative, wherein the cellulose derivative is hydroxypropylcellulose, hydroxypropyl methylcellulose, or the mixture thereof, dissolved in a polar solvent; wherein the preferred polar solvent used to dissolve the carrier is water; wherein the temperature of the solution of the carrier during the addition of the dissolved turmeric extract falls within the temperature from about 5 °C to about 10 °C, wherein preferably cooling, maintaining this temperature of the resulting mixture, is applied;

wherein, preferably, the mixture of hydroxypropylcellulose to hydroxypropylmethylcellulose is in a ratio of 10:1;

wherein, preferably, the carrier solution is obtained by mechanically mixing the polymers, wherein the mixing is preferably carried out for not less than 10 hours, preferably about 12 hours;

wherein, preferably, the carrier solution is dissolved in a polar solvent in a weight ratio of 1:5 to 1:25, preferably 1:10;

c) the mixture obtained in step b) is homogenized to obtain an emulsion, by homogenization at the temperature from about 5 °C to about 10 °C;

d) the emulsion obtained in step c) is diluted with an aqueous solution of β-glucan with a low viscosity and is further briefly homogenized; preferably, with the viscosity in the range of 40 – 200 mPas,

preferably the emulsion is diluted with 1-8% β-glucan, preferably with β-glucan from oats or barley, or the mixture thereof;

e) the emulsion obtained in step d) is stabilized by dilution with an aqueous solution of a cellulose derivative with a high viscosity, above 5600 mPas, diluting in a ratio of not less than 1:6, preferably not less than 1:8, stirring it all mechanically,

wherein, preferably, the cellulose derivative with high viscosity is carboxymethyl cellulose (CMC), preferably in the concentration from 0.01 to 0.5%, more preferably in the concentration from 0.10 to 0.20%, more preferably in the concentration from 0.1 to 1,4% w/w, and

f) the stabilized emulsion from step e) is dried to provide the curcumin formulation in a dry form,

wherein over 90%, preferably over 95%, preferably over 99% of the obtained particles have a diameter size of below about 1 µm,

more preferably at least 90%, more preferably at least 95% of the particles, more preferably at least 99% of the particles have a diameter in the range from 0.1 to about 1 µm,

wherein preferably drying is carried out using spray drying with the temperature

at the inlet of the dryer in the range of 110 – 160°C, more preferably about 150°C, and

not more than 65°C at the outlet, preferably not more than about 55°C.

5. A curcumin formulation in a dry form produced by the method as defined in claim 4.

6. A beverage concentrate comprising the formulation as defined in claims 1-3 and/or 5.

7. The beverage concentrate according to claim 6, wherein it further comprises at least one of the following:

- a stabilizer and thicker in the concentration of 0-28% w/w, preferably in the concentration of 0-20% w/w, preferably glycerol and/or xanthan gum;

- an antioxidant and acidity regulator 0-5% w/w, preferably ascorbic acid;

- an acidity regulator 0-5% w/w, preferably citric acid;

- an electrolyte and sweetener 0-80% w/w, preferably glucose;

- B vitamins, preferably selected from thiamine, niacin, pyridoxine, folic acid and riboflavin;

- preservatives, preferably benzoic acid, sodium benzoate, potassium sorbate, or the mixture thereof.

8. A beverage comprising the formulation as defined in claims 1-3 and/or 5 or obtained by diluting the concentrate defined in claims 6-7.

9. The beverage according to claim 8, wherein it is the beverage based on fruit juices, water, milk beverage, plant beverage substituting milk.

10. The beverage according to claims 8-9, wherein it is a functional beverage supporting liver function, wherein preferably, the concentration of curcuminoids in the beverage is 0.01-0.1% w/w.

11. A dietary supplement comprising the formulation as defined in claims 1-3 and/or 5, preferably in a liquid form, capsules, tablets, orally approved forms.

12. The dietary supplement according to claim 11, wherein it supports liver function, wherein, preferably, it activates the action of liver enzymes, preferably the enzymes of the dehydrogenases group, preferably aldehyde dehydrogenase.

13. The dietary supplement according to claim 12, wherein it supports the anti-inflammatory effect and is intended for anti-inflammatory prevention and reduction of inflammation including the prevention of complications and severe course of viral infections, coronavirus infections, SARS-CoV-2 infection, preferably for use in the period of increased viral infections incidence.

14. The dietary supplement according to claims 11-13, characterized in that it is in the liquid form and comprises

-formulation as defined in claims 1-3 and/or 5 in the amount providing curcuminoids at a concentration of 0.01-0.1% w/v,

- fruit juice concentrate 1-20% w/v,
- sweetener 2-10% w/v,
- stabilizing and thickening agent in a concentration of 2-10% w/v.

15. A pharmaceutical composition comprising the formulation as defined in claims 1-3 and/or 5 and a pharmaceutically acceptable additive.

16. A pharmaceutical composition comprising the formulation as defined in claims 1-3 and/or 5 for use as a medicine.

17. The pharmaceutical composition for use according to claim 16, wherein it is for use as the medicine in the regulation of liver function, medicine with an anti-inflammatory, anticancer and antidepressant activity.

18. The pharmaceutical composition for use according to claim 16 -17, wherein it is for use as the medicine in the regulation of liver function, activating the action of liver enzymes, preferably the enzymes of the dehydrogenases group, preferably aldehyde dehydrogenase.

19. A pharmaceutical composition comprising the formulation as defined in claims 1-3 and/or 5 for use as a liver function stabilizer, an agent supporting the liver, an agent activating the action of liver enzymes, preferably the enzymes of the dehydrogenases group, preferably aldehyde dehydrogenase.

20. A pharmaceutical composition comprising the formulation as defined in claims 1-3 and/or 5 for use as an agent supporting the anti-inflammatory effect intended for anti-inflammatory prevention and reduction of inflammation, preferably the prevention of complications and severe course of viral infections, including coronavirus infections, including SARS-CoV-2, wherein it is preferably used in the period of increased viral infections incidence.

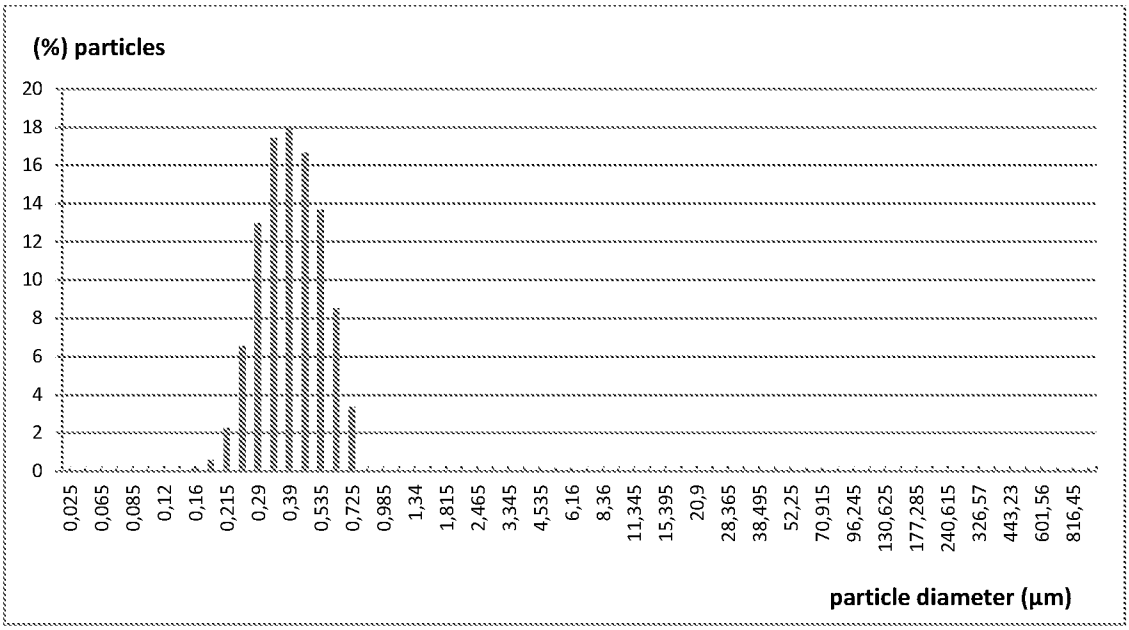
21. A dye containing the formulation as defined in claims 1-3 and/or 5.

22. The dye according to claim 21, characterized in that it is a food dye used in medicine, cosmetics, food, animal feed.

23. Use of the curcumin formulation as defined in claims 1-3 and/or 5 as a functional additive to the beverage, as a dye, as an active ingredient in the dietary supplement, as an active ingredient in the pharmaceutical composition, as a food additive, as a feed additive.

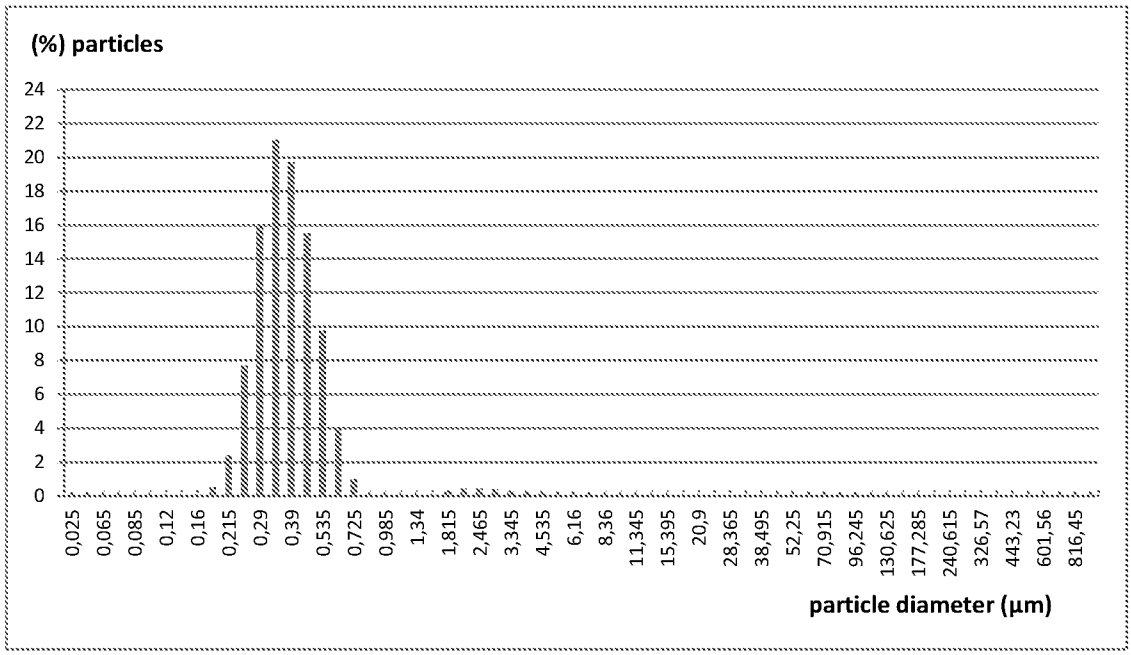
24. Use of the curcumin formulation as defined in claims 1-3 and/or 5 as an anti-hangover agent, an agent for prevention and/or alleviation -hangover symptoms or as an agent activating liver enzymes, preferably the enzymes of the dehydrogenases group, preferably aldehyde dehydrogenase.

FIG. 1



Percentage distribution of particle sizes in the sample (emulsion) in the particular size categories.

FIG. 2



Percentage distribution of particle sizes in the sample (powder after dissolution), in particular size categories.

INTERNATIONAL SEARCH REPORT

International application No.
PCT/PL2021/050009

A. CLASSIFICATION OF SUBJECT MATTER

A61K9/14, A61K36/9066, A61K47/36, A61K47/38, A61K47/54, A61K47/61, A23L 2/00, A23L33/105, A61P1/16, A61P31/12, A61P29/00, A23K20/00, A23L5/43, C09B 61/00

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

Epodoc&WPI via Epoquenet, Database of PPO, Internet (Google)

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
A	WO2009007273 A2; CHR HANSEN AS (DK); 2009-01-15	1-16, 22-23
A	WO2014191556 A1; Chr. Hansen A/S, Denmark (DK); 2014-12-04	1-14, 21-23
A	CN106579421 A; UNIV SOUTH CHINA TECH; 2017-04-26	1-20, 23
A	EP3120843 A1; THERAVALUES CORP [JP]; TAKEUCHI HIROFUMI (JP); 2017-01-25	1-5
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International application No.
PCT/PL2021/050009

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